

Towards Foundation Models for MMKG: Multi-Task Inductive Generalization via Task-Aware Routing

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Abstract

Multimodal knowledge graphs (MMKG) extend the representational capacity of conventional knowledge graphs by integrating multimodal contexts. However, despite increasing attention on core tasks—multimodal link prediction (MMLP), multimodal entity alignment (MMEA), and multimodal entity linking (MMEL)—most existing approaches remain confined to meticulously tailored, task-specific and graph-specific paradigms, limiting their practicality in dynamic environments where tasks co-exist and new knowledge graphs continually emerge. In this work, we propose the first unified **multi-task inductive inference framework for MMKG (MtaIMKG)** that can handle multiple core tasks, with **universality** and **transferability** to any unseen MMKG, by tackling inherent heterogeneities: i) Feature Heterogeneity: Bridging the semantic gaps in feature representations across domains. ii) Structural Heterogeneity: Overcoming structural divergence in topology across graphs. iii) Task Heterogeneity: Reconciling conflicting objectives and inductive biases across tasks. To address the above challenges, we center on reconciling divergent representation preferences across tasks, delivering only task-pertinent information instead of engaging in indiscriminate aggregation. Specifically, we introduce a task-aware routing network that dynamically integrates and routes multimodal representations, assesses their contributions under task semantics, and produces adaptive, task-conditioned representations for any

input graph. In addition, MtaIMKG employs a gated mixture-of-experts to suppress cross-modal noise and extract complementary information across tasks. These strategies tackle the notable heterogeneity challenges in multi-task inductive inference on MMKG, enabling MtaIMKG to effectively zero-shot generalize to any unseen graph. Extensive experiments demonstrate that MtaIMKG achieves SOTA performance in both multi-task inductive and transductive inference with competitive efficiency, confirming its value as a scalable and generalizable solution for MMKG reasoning. Our code is released at <https://github.com/MMKGer/MtaIMKG/>.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

Keywords

Multi-Task Inductive Learning, Multimodal Knowledge Graph, Knowledge Representation Learning and Embedding

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1 INTRODUCTION

Knowledge graphs (KG) model [1, 35, 37] real-world knowledge as triples comprising a head entity, a relation, and a tail entity (h, r, t) . With the explosive growth of multimodal data such as text and images, traditional KG have evolved into multimodal knowledge graphs (MMKG) [46, 51, 52], which attach multiple modality-specific attributes to each entity to enrich its semantic representation. Such

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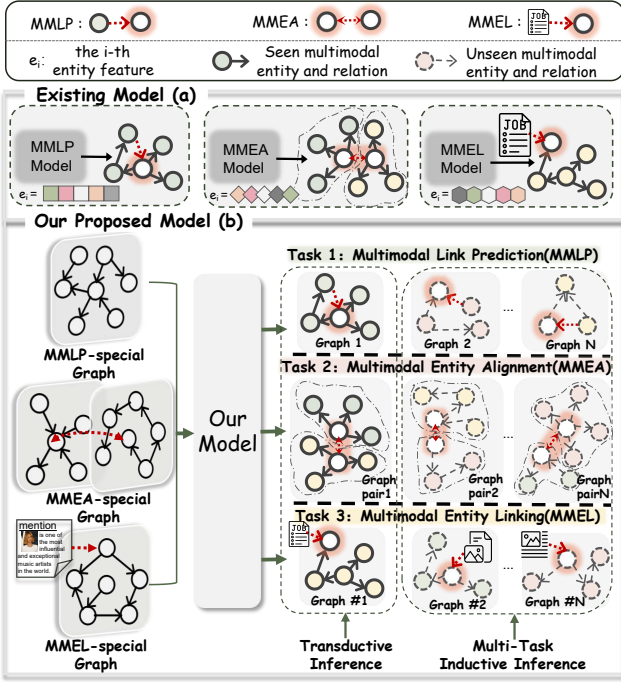


Figure 1: Conventional task-specific models (top) vs. Our proposed multi-task inductive framework (bottom).

enriched representations have demonstrated significant practical value in core Web applications including recommender systems[8, 25], web search[11, 32, 33], and intelligent question answering[15, 57], as well as in specialized domains like healthcare[31, 39]. The Web’s inherently multimodal and interconnected nature makes MMKG particularly suitable for addressing fundamental challenges in web-scale data integration and understanding. Driven by these applications, several core tasks have emerged: multimodal link prediction (MMLP)[46, 47], which infers plausible missing triples to complete the graph; multimodal entity alignment (MMEA)[20, 22], which identifies semantically equivalent entities across different MMKG; and multimodal entity linking (MMEL)[13, 17], which matches mentions from text or images to their candidate entities in the graph.

Despite preliminary advances in MMKG tasks, existing methods [21, 34, 46] remain largely confined to tailored learning frameworks meticulously designed for specific tasks and particular graphs. As illustrated in Fig. 1(a), models developed for tasks such as MMLP, MMEA, and MMEL are typically trained in isolation and tightly coupled with individual datasets. This highly specialized paradigm presents significant challenges: when new knowledge graphs emerge, these models often require extensive retraining for adaptation, which is not only computationally expensive and resource-intensive but also time-consuming. More critically, when limited data is available for tasks, the potential to leverage historical knowledge for enhanced performance is severely constrained. This siloed paradigm severely limits their applicability in dynamic real-world environments, where multiple tasks often coexist and new knowledge graphs emerge continuously. To overcome the limitations of existing MMKG models and accommodate dynamic real-world

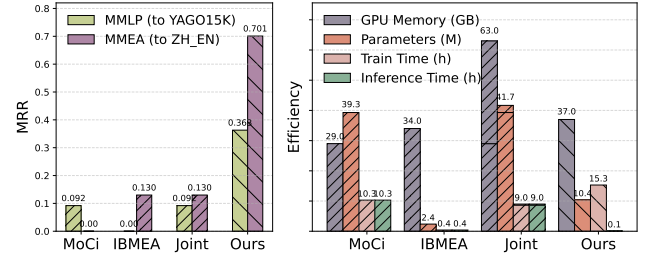


Figure 2: Performance-Efficiency Trade-off: (Left) Superior inductive performance on MMLP/MMEA; (Right) Efficient resource utilization with fast adaptation.

scenarios, we introduce a novel multi-task[5] inductive inference paradigm for MMKG (Fig. 1(b)). This framework enables a unified model to jointly address multiple MMKG tasks and perform inductive reasoning over completely unseen graphs, thereby facilitating cross-task knowledge sharing and cross-graph generalization.

However, compared to conventional task-specialized MMKG models, the multi-task inductive reasoning paradigm for MMKG introduces a series of complex challenges rooted in the inherent heterogeneity of multimodal knowledge graphs[41, 42]. **Feature heterogeneity**: Cross-graph scenarios exhibit substantial heterogeneity in node and edge features, especially in multimodal content, as well as significant cross-dataset disparities in feature dimensionality and semantics, collectively hindering effective cross-domain generalization. **Structural heterogeneity**: Domain-specific knowledge graphs exhibit substantial divergence in topological characteristics, such as average degree, frequency of dense subgraphs, missing-edge ratio, and the prevalence of multi-hop dependencies and path distributions. These fundamental differences cause propagation or positional encoding mechanisms learned on one graph to fail to generalize to new domains, ultimately resulting in significant inductive performance degradation. **Task heterogeneity**: MMLP focuses on completing missing triples, MMEA aims to identify equivalent entities across graphs, and MMEL deals with linking external mentions to candidate entities. These tasks not only differ in objectives but also often entail conflicting inductive biases and modeling granularities. Consequently, a unified model must align heterogeneous features, bridge cross-graph distribution gaps, robustly capture complementary multimodal information across tasks, and maintain strong generalization to completely unseen domains and graphs, all without relying on task, domain, or graph specific customization. Fig. 2 (Left) presents the performance of current task-specialized models in multi-task inductive settings, revealing that even leading models struggle with effective cross-graph inductive inference.

To address the above challenges, we reconcile divergent representation preferences across tasks and deliver only task-pertinent information rather than engaging in indiscriminate aggregation by introducing a novel task-aware routing network that dynamically integrates and routes multimodal representations to generate adaptive, task-conditioned embeddings for any input graph. Specifically, we extract meta-relational neighborhoods into a universal structural representation to ensure robustness across diverse topologies, while learning transferable textual and visual representations that suppress inter-modal noise and extract complementary

information prior to routing. Furthermore, we employ a mixture-of-representations router that assesses each modality’s contribution under task semantics and performs fine-grained, channel-wise selection to emphasize task-critical features. Finally, lightweight task heads perform decoupled inference on the routed multimodal representations. Fig. 1 compares conventional MMKG models with our proposed multi-task inductive reasoning framework. Unlike existing models that rely on task-specific and graph-specific embedding learning, our approach generates adaptive, task-conditioned embeddings for any input graph. Furthermore, Fig. 2 demonstrates that our model achieves superior multi-task inductive performance while maintaining competitive efficiency. This design enables transferable representation learning and supports effective cross-task and cross-graph inductive inference within MMKG. As the first unified framework for multi-task inductive reasoning on MMKG, our method provides a scalable and generalizable solution for cross-task transfer, cross-graph reasoning, and cross-modal interaction. Our contributions are as follows:

- Our MtaIMKG is the first unified framework for multi-task inductive inference on MMKG. It marks a paradigm shift by seamlessly unifying MMLP, MMEA, and MMEL within a single model that achieves zero-shot generalization to unseen graphs, moving beyond task-specialized models.
- We propose a novel task-aware routing network that dynamically integrates and routes multimodal representations to generate adaptive, task-conditioned embeddings for any input graph, effectively addressing feature, structural, and task heterogeneity in multimodal knowledge graphs.
- Our model achieves superior multi-task inductive inference, surpassing the best specialized baselines by up to 329% on MMEA and 71% on MMLP while maintaining strong MMEL performance, all achieved with competitive efficiency. Notably, MtaIMKG also achieves SOTA transductive link prediction, improving MRR by 5.1% with only 10.4M parameters, $3.8\times$ fewer than specialized models.

2 RELATED WORK

2.1 Multimodal Link Prediction

With the emergence of MMKG, MMLP has gained increasing attention. Models such as IKRL[43], AdaMF[53], and NativE[50] focus on integrating multimodal information by projecting modality-specific features into a unified embedding space. Other approaches including MKGformer[6], MULTIFORM[49], and TBKGC[28] further explore cross-modal interactions, though often at the cost of preserving distinct modal characteristics. To address this, IMF[21] emphasizes independent learning of modality-specific features, while MoCi[46] captures and integrates inter-entity modal semantics. More recent methods like the fine-grained multimodal tokenization approach MyGO[52] and MoMoK[51] employ mixture-of-experts networks to disentangle modal representations for improved link prediction. However, existing MMLP models are typically tailored for specific graphs and tasks, limiting their generalization to new domains and hindering application in dynamic multi-task scenarios.

2.2 Multimodal Entity Alignment

With the emergence of MMKG, MMEA has gained significant attention for improving alignment accuracy through multimodal integration. Early methods such as MEAformer[7] incorporate transformers to dynamically learn modality weights, while MCLEA[22] leverages contrastive learning to enhance cross-modal matching. To address low-resource scenarios, SimDiff[20] introduces diffusion-based data augmentation to stabilize training. Further advances include IBMEA[34], which applies the information bottleneck principle to filter out misleading multimodal cues, alongside other methods that integrate multimodal information through various fusion mechanisms. Despite these advancements, existing MMEA methods remain constrained by their reliance on supervised data from specific graph pairs and predefined alignment settings, limiting their adaptability to unseen graphs or multi-task environments.

2.3 Multimodal Entity Linking

With the emergence of MMKG, MMEL extends traditional entity linking by incorporating multimodal information to resolve entity ambiguity. Models such as vision-language pre-training methods leverage multimodal data to align semantic representations[9, 16, 19, 29]. Other approaches include generative-based methods that utilize large language models with constrained decoding to generate entity candidates[24, 27, 30]. Additionally, multimodal interaction methods fuse entity and mention representations through attention, matching networks, or optimal transport[13, 17, 55]. However, these methods are typically designed for specific datasets and struggle to generalize across different KG domains or adapt to multi-task scenarios, limiting their applicability in dynamic environments.

3 TASK FORMULATION

A multi-task inference setting over MMKG, defined by the task set $\mathcal{T}_{\text{task}} = \{\text{MMLP}, \text{MMEA}, \text{MMEL}\}$. An MMKG is formally represented as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{F})$, where $\mathcal{E}$ denotes the entity set, $\mathcal{R}$ the relation set, and $\mathcal{F} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$ the set of factual triples. Each entity $e \in \mathcal{E}$ is associated with multimodal information from structural (s), visual (v), and textual (t) modalities, collectively $M = \{s, v, t\}$.

Multimodal Link Prediction (MMLP). Given an incomplete triple $(h, r, ?)$ or $(?, r, t)$, MMLP infers missing entities through integration of multimodal contexts $\{e_h^m, e_t^m\}_{m \in M}$ with relation representation r , enhancing knowledge graph completeness.

Multimodal Entity Alignment (MMEA). Given two MMKG $\mathcal{G}^A = (\mathcal{E}^A, \mathcal{R}^A, \mathcal{F}^A)$ and $\mathcal{G}^B = (\mathcal{E}^B, \mathcal{R}^B, \mathcal{F}^B)$ with seed alignments $\mathcal{A} \subseteq \mathcal{E}^A \times \mathcal{E}^B$, MMEA identifies new entity matches by integrating graph structures and modal feature similarities $\{(e_a^m, e_b^m)\}_{m \in M}$.

Multimodal Entity Linking (MMEL). Given a mention u with multimodal context and a candidate set $\mathcal{E}_u \subseteq \mathcal{E}$, MMEL identifies the true entity $y \in \mathcal{E}_u$ by matching the multimodal features of the mention to those of candidate entities $\{e^m\}_{m \in M}$.

Multi-Task Inductive Inference. We introduce a novel and highly challenging multi-task inductive inference regime for MMKG: a unified model jointly trained on $\mathcal{T}$ that generalizes to completely unseen graphs $\mathcal{G}_\tau^{\text{inf}}$ under the constraints: $\mathcal{E}_\tau^{\text{train}} \cap \mathcal{E}_\tau^{\text{inf}} = \emptyset$, $\mathcal{R}_\tau^{\text{train}} \cap \mathcal{R}_\tau^{\text{inf}} = \emptyset$, $\mathcal{G}_\tau^{\text{train}} \cap \mathcal{G}_\tau^{\text{inf}} = \emptyset$. In this paradigm, the model must infer adaptive representations for new entities and relations based solely on transferable parameters, multimodal attributes, and local graph

structure, while simultaneously achieving cross-graph zero-shot generalization and facilitating knowledge transfer across heterogeneous objectives. Although focused on the fully inductive setting, the proposed approach naturally accommodates conventional transductive and partially inductive scenarios for each task.

4 METHODOLOGY

Existing MMKG models are meticulously tailored to specific tasks and graphs, which prevents them from handling structural, feature, and task heterogeneity and from scaling to multi-task settings, ultimately limiting generalization. To address this, we propose a unified framework (Fig. 3). Our framework captures universal topological structures through meta-relational propagation and learns transferable textual and visual representations via Mixture-of-Experts Cross-Modal Learning, effectively suppressing cross-modal noise while extracting complementary information. Furthermore, it incorporates a task-aware multimodal router that dynamically assesses each modality's contributions to emphasize task-critical features based on task semantics, while lightweight task heads enable efficient and decoupled inference across diverse tasks.

4.1 Graph-Agnostic Structural Learning

Existing methods are typically tailored to specific tasks and graphs, tightly coupling structural representations to a particular dataset and training graph (e.g., average degree, path patterns), which fundamentally limits inductive generalization across graph distributions. To address this limitation, we propose Graph-Agnostic Structural Learning inspired by [58], which exploits the intrinsic relational structure of knowledge graphs (meta-relations), captures cross-task and cross-graph topological regularities, and extracts structural knowledge agnostic to any particular task or graph.

Specifically, given relations $\mathcal{R}$ and four meta-relation types $\mathcal{P} = \{h-h, h-t, t-h, t-t\}$ (head-head, head-tail, tail-head, tail-tail), we build the meta-relation graph $\mathcal{G}_r = (\mathcal{R}, \mathcal{P}, \mathcal{T}_p)$ with edges $(r_1, p, r_2) \in \mathcal{T}_p$. For a given task τ and at each layer l , the updates for relation and entity structural representations are defined as:

$$\begin{aligned} \mathbf{r}_\tau^{l+1} &= \mathbf{W}_r \left[\mathbf{r}_\tau^l; A_{\text{sum}} \left(M_{dm}(\mathbf{r}_{w|\tau}^l, \mathbf{p}) \mid w \in \mathcal{N}_p(r_\tau), p \in \mathcal{P} \right) \right], \\ \mathbf{e}_\tau^{l+1} &= \mathbf{W}_e \left[\mathbf{e}_\tau^l; A_{\text{sum}} \left(M_{dm}(\mathbf{e}_{w|\tau}^l, g(\tilde{\mathbf{r}}_\tau)) \mid (w, r_\tau) \in \mathcal{N}_r(e_\tau) \right) \right], \end{aligned} \quad (1)$$

where $\mathbf{W}_r, \mathbf{W}_e \in \mathbb{R}^{2d \times d}$ are projection matrices, $M_{dm}(\mathbf{a}, \mathbf{b}) = \mathbf{a} \odot \mathbf{b}$ is a parameter-free DistMult interaction function, $A_{\text{sum}}(\cdot)$ denotes sum aggregation, and $g(\cdot)$ is a task-conditioned adapter implemented as an MLP. $\mathcal{N}_p(r)$ refers to the neighbors of relation r under meta-relation type p in $\mathcal{G}_r$, while $\mathcal{N}_r(e)$ refers to the neighbors of entity e under relation r in the original graph $\mathcal{G}$. Initialized based on the input structure: for MMLP with query $(h, r, ?)$, set $\mathbf{r}_\tau^1 = \mathbf{1}_{r=r} \cdot \mathbf{1}^d$ and $\tilde{\mathbf{r}}_\tau = \mathbf{r}_\tau^1$; for MMEA/MMEL, define $\tilde{\mathbf{r}}_\tau$ by relation-wise averaging $\tilde{\mathbf{r}}_\tau = \frac{1}{|\mathcal{R}|} \sum_k \mathbf{r}_{k|\tau}^1$. The structural representations for entity e is defined as $\mathbf{e}_\tau^s = \mathbf{e}_\tau^l$. This propagation mechanism is graph-agnostic, capturing transferable meta-relations inherent in any knowledge graph. By incorporating task information solely through the context representation $\tilde{\mathbf{r}}_\tau$, the same architecture effectively supports MMLP, MMEA, and MMEL in inductive settings, enabling reasoning over unseen entities and relations in new knowledge graphs.

4.2 Mixture-of-Experts Cross-Modal Learning

Although multimodal information enriches entity semantics, feature heterogeneity from cross-domain divergences in semantic and feature spaces poses significant challenges in cross-task and cross-graph settings. To address this, we introduce a gated mixture-of-experts within a decoupled framework that learns transferable textual and visual representations while suppressing cross-modal noise and extracting complementary information.

MtaIMKG follows the standard feature extraction methods of each dataset to obtain raw textual and visual features, employing a shallow transformer encoder to project these heterogeneous features into a unified $\mathbb{R}^d$ semantic space and producing initial embeddings $\mathbf{e}^t, \mathbf{e}^v$. To consolidate complementary information while attenuating cross-modal noise, a gated mixture-of-experts operates on the base representation $\frac{1}{2}(\mathbf{e}^t + \mathbf{e}^v)$: a gating network $\mathbf{g} = \text{softmax}(\mathbf{W}_{g_1} \sigma(\mathbf{W}_{g_1} \mathcal{N}([\mathbf{e}^t \parallel \mathbf{e}^v])))$ with $\mathbf{W}_{g_1} \in \mathbb{R}^{h \times 2d}$, $\mathbf{W}_{g_2} \in \mathbb{R}^{K \times h}$ assigns mixture weights over K experts, where each $\text{Expert}_k(\cdot)$ is implemented as an MLP: $\text{Expert}_k(\mathbf{x}) = \mathbf{W}_{k_2} \sigma(\mathbf{W}_{k_1} \mathcal{N}(\mathbf{x}))$ with $\mathbf{W}_{k_1} \in \mathbb{R}^{h \times d}$, $\mathbf{W}_{k_2} \in \mathbb{R}^{d \times h}$. The fused cross-modal representation:

$$\mathbf{e}_f = \sum_{k=1}^K g_k \cdot \text{Expert}_k\left(\frac{1}{2}(\mathbf{e}^t + \mathbf{e}^v)\right), \quad (2)$$

which is further injected into each modality $m \in \{t, v\}$ through a linear transformation $\mathbf{P}^m \in \mathbb{R}^{d \times d}$ to establish a task-shared interaction. Formally, for entity e , task τ , and modality m , robustly extracting complementary information, the task-conditioned embedding:

$$\mathbf{e}_\tau^m = \text{Ada}_\tau^m \left(\text{Film}_\tau^m(\mathbf{e}_\tau^m + \mathbf{P}^m \mathbf{e}_f) \right), \quad (3)$$

where $\text{Film}_\tau^m(\mathbf{x}) = \mathbf{x} \odot (1 + \gamma_\tau^m) + \beta_\tau^m$ with $[\gamma_\tau^m, \beta_\tau^m] = \mathbf{W}_{\text{film}}^m(\mathcal{N}(\mathbf{e}_\tau))$ derived from the task embedding $\mathbf{e}_\tau \in \mathbb{R}^d$; the low-rank adapter $\text{Ada}_\tau^m(\mathbf{x}) = \mathbf{x} + \mathbf{U}_\tau^m \phi(\mathbf{V}_\tau^m \mathcal{N}(\mathbf{x}))$ employs compressed parameter matrices $\mathbf{V}_\tau^m \in \mathbb{R}^{r \times d}$, $\mathbf{U}_\tau^m \in \mathbb{R}^{d \times r}$ with adapter rank $r < d$; $\phi(\cdot)$ is a pointwise nonlinearity, $\sigma(\cdot)$ denotes the GELU activation function, and $\mathcal{N}(\cdot)$ represents a normalization operation LayerNorm. The resulting task-aware representations are characterized by transferability, noise suppression, and cross-graph compatibility, while incorporating task-specific calibration for effective routing. These calibrated textual and visual representations provide optimized inputs to the mixture-of-representations router.

4.3 Task-Aware Multimodal Router

While our learned non-specific graph, general-purpose multimodal representations provide a solid foundation for cross-graph reasoning, they remain insufficiently task-specific to accommodate the pronounced differences in objectives and inductive biases across tasks. To bridge this gap, we introduce a mixture-of-representations router that dynamically assesses each modality's contributions based on task semantics and performs fine-grained, channel-wise selection within each modality to emphasize task-critical features.

Specifically, the routing mechanism takes as input the refined, bias-corrected representations from preceding modules, which retain only task-essential, cross-graph generalizable features and collectively provide the mixture components for routing. For entity e and task τ with $M = \{s, t, v\}$, we build a token sequence $[\{\mathbf{e}_\tau^m\}_{m \in M}, \mathbf{e}_\tau]$ and feed it to a shallow transformer layer to obtain modality states $\tilde{\mathbf{e}}_\tau^m$ and a task state $\mathbf{e}_\tau$. The router computes modality-level weights $\alpha_m(e, \tau) = \text{softmax}_m(\langle \tilde{\mathbf{e}}_\tau^m, \mathbf{e}_\tau \rangle / \kappa)$, where

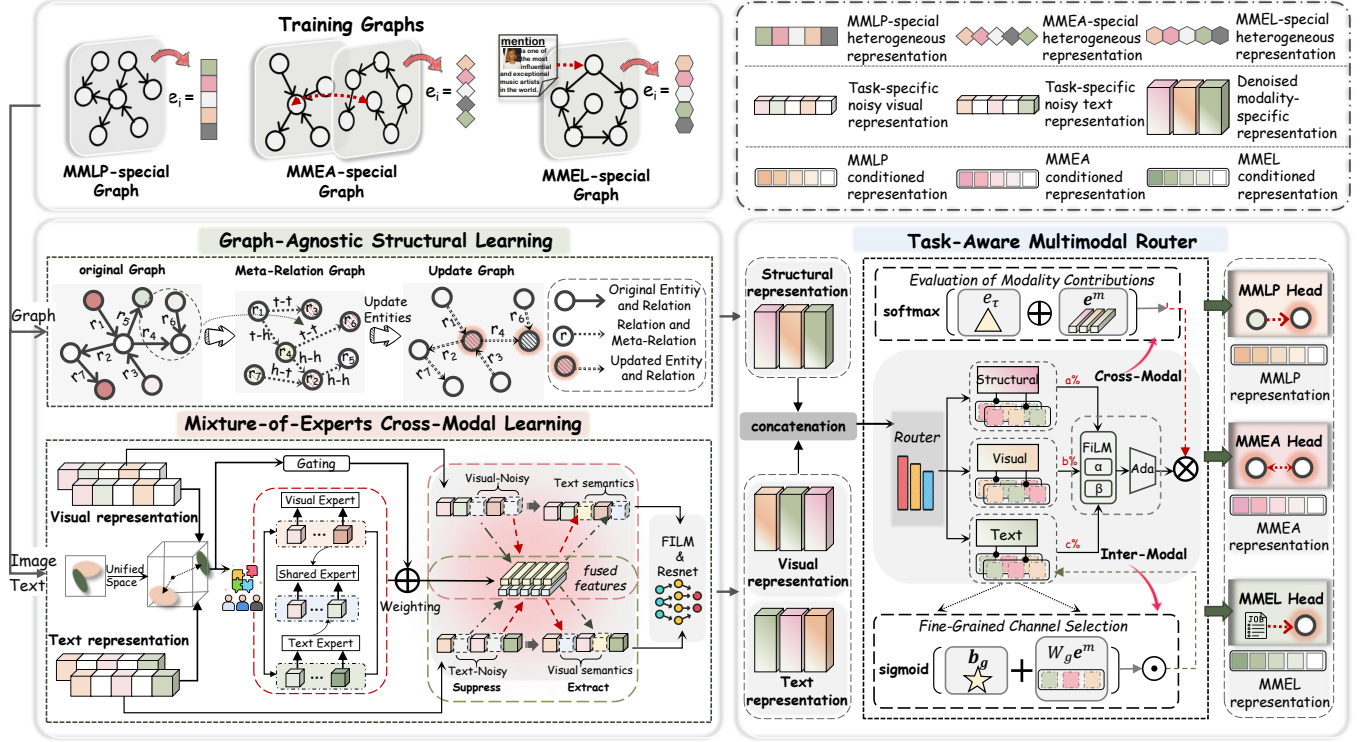


Figure 3: Overview of the MtaIMKG model. MtaIMKG employs Graph-Agnostic Structural Learning (Sec. 4.1) to capture universal topological structures. Mixture-of-Experts Cross-Modal Learning (Sec. 4.2) for learning transferable textual/visual representations. Task-Aware Multimodal Router (Sec. 4.3) for emphasizing task-critical features and task inference.

$\kappa > 0$ is a temperature parameter, together with channel-wise gates $\mathbf{g}^m(e, \tau) = \sigma(\mathbf{W}_g \tilde{\mathbf{e}}^m + \mathbf{b}_g) \in (0, 1)^d$, with $\sigma(\cdot)$ denoting the sigmoid function, producing the routed multimodal representation:

$$\mathbf{e}_{\text{router}|\tau}^m = \sum_{m \in M} \alpha_m(e, \tau) \text{Ada}_\tau^m(\text{FiLM}_\tau^m(\mathbf{e}_\tau^m \odot \mathbf{g}^m(e, \tau))), \quad (4)$$

where $\odot$ denotes element-wise multiplication, $\text{FiLM}_\tau^m(\cdot)$ applies task-conditioned feature-wise linear modulation, and $\text{Ada}_\tau^m(\cdot)$ is the low-rank adapter for lightweight task-specific calibration. All router parameters are shared across tasks to facilitate knowledge transfer. The resulting routed entity representation $\mathbf{e}_{\text{router}|\tau}^m$ is directly fed to the task-specific heads, which perform decoupled inference for MMLP, MMEA, and MMEL.

Multimodal Fusion for Multimodal Link Prediction. We introduce a lightweight mixture of experts with multiple experts $\{\text{Expert}_k\}_{k=1}^K$ to generate content-adaptive fused embeddings compatible with any link prediction scorer. A shared projection layer Φ_{align} first maps all modalities into a common space, and the fused entity representation is obtained via expert attention gating:

$$\mathbf{z} = \sum_{k=1}^K \alpha_k \cdot \text{Expert}_k(\rho_k(\Phi_{\text{align}}(\mathbf{e}_\tau^t), \Phi_{\text{align}}(\mathbf{e}_\tau^v), \Phi_{\text{align}}(\mathbf{e}_\tau^s))), \quad (5)$$

where $\Phi_{\text{align}}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is implemented as a two-layer MLP with LayerNorm and dropout, ρ_k selects the appropriate modality subset for expert k , and α_k denotes the attention weight produced by the router network. The triple scoring function combines structural and multimodal pathways: $s(h, r, t) = \frac{1}{2} f_{lp}^s(\mathbf{e}_{h|t}^s, \mathbf{e}_{t|t}^s) + \frac{1}{2} f_{lp}^z(\mathbf{z}_h, \mathbf{z}_t)$, where f_{lp} can be any differentiable link prediction scorer.

Gated Fusion for Multimodal Entity Alignment. We implement a lightweight refinement layer that performs gated fusion and graph propagation over routed multimodal representations. The module employs a learned gate network $\mathbf{g}$ to generate modality weights ω , followed by weighted concatenation and linear projection via $\mathbf{Q} \in \mathbb{R}^{d' \times d_{\text{concat}}}$. A graph attention network propagates the fused representation over the adjacency matrix $\mathbf{A}$ through L iterative refinement rounds:

$$\mathbf{z}^{(l+1)} = \mathbf{z}^{(l)} \cup \{\text{GAT}(\mathbf{Q} \cdot \oplus_{m \in \mathbf{z}^{(l)}} [\omega]_m \hat{\mathbf{e}}^m, \mathbf{A})\}, \quad (6)$$

where $\omega = \text{softmax}(\mathbf{g}(\hat{\mathbf{e}}^m \in \mathbf{z}^{(l)}))$, $\hat{\mathbf{e}}^m$ are normalized embeddings, $\mathbf{z}^{(0)}$ initializes with text/image features, and $\oplus$ denotes the concatenation operation. The fused representation $\mathbf{z}^*$ from $\mathbf{z}^{(L)}$ computes alignment scores via $s(u, v) = f_{\text{ea}}(\mathbf{z}_u^*, \mathbf{z}_v^*)$.

Cross-Modal Integration for Multimodal Entity Linking. We develop a lightweight bilinear matcher with low-rank residuals for adaptive cross-modal entity linking. For mention u with candidates $\mathcal{E}$, we integrate structure into candidates: $\mathbf{E}^t = \mathbf{W}^t[\mathbf{e}^t; \tilde{\mathbf{e}}^s] + \mathbf{b}^t$ and $\mathbf{E}^v = \mathbf{W}^v[\mathbf{e}^v; \tilde{\mathbf{e}}^s] + \mathbf{b}^v$, where $\tilde{\mathbf{e}}^s$ is the normalized structural representation. We further apply gating and normalization: $\tilde{\mathbf{u}}^t = \mathcal{N}(\mathbf{u}^t \odot \tanh(\mathbf{W}_g \mathbf{u}^t))$, $\tilde{\mathbf{E}}^t = \mathcal{N}(\mathbf{E}^t \odot \tanh(\mathbf{W}_g \mathbf{E}^t))$, similarly for visual modality, where $\mathbf{W}_g \in \mathbb{R}^{d \times d}$ is a shared gate projection. The router matching score incorporates low-rank bilinear residuals:

$$f_{el}^{\text{router}}(u, \mathcal{E}) = \frac{1}{2} \sum_{m \in \{t, v\}} \text{softmax}_{\mathcal{E}}(\tilde{\mathbf{u}}^m (\tilde{\mathbf{E}}^m)^\top + \mathbf{B}_m(\mathbf{u}^m, \mathbf{E}^m)). \quad (7)$$

where $\mathbf{u}^t, \mathbf{u}^v \in \mathbb{R}^d$ and $\mathbf{E}^t, \mathbf{E}^v \in \mathbb{R}^{|\mathcal{E}| \times d}$ represent mention and candidate embeddings, with $\mathbf{B}_{\{\cdot\}}$ denoting low-rank bilinear terms. The matching scoring function combines textual, visual, cross-modal, and router pathways $s(u, \mathcal{E}) = f_{el}^v(u, \mathcal{E}) + f_{el}^t(u, \mathcal{E}) + f_{el}^{vt}(u, \mathcal{E}) + f_{el}^{router}(u, \mathcal{E})$, where f_{el} can be any differentiable matching scorer.

Training. Our model is trained in a multi-task learning framework, with the total loss defined as a weighted sum of the task-specific losses for MMLP, MMEA, and MMEL:

$$\mathcal{L} = \lambda_{lp} \mathcal{L}_{lp} + \lambda_{ea} \mathcal{L}_{ea} + \lambda_{el} \mathcal{L}_{el}, \quad (8)$$

where λ_{lp} , λ_{ea} , and λ_{el} are balancing hyperparameters, which enable dynamic adjustment of each task’s contribution during training for stable optimization and enhanced knowledge transfer.

Table 1: Overview of Datasets.

Task	Dataset	Entities	Relations	Triples
MMLP	YAGO15K	15,283	32	122,886
	DB15K	14,777	279	99,028
	FB15K-237	14,541	237	310,116
	MKG-W	15,000	169	42,746
	MKG-Y	15,000	28	26,638
MMEA	DBP15K _{ZH-EN} (ZH)	19,388	1,701	70,414
	DBP15K _{ZH-EN} (EN)	19,572	1,323	95,142
	DBP15K _{JA-EN} (JA)	19,814	1,299	77,214
	DBP15K _{JA-EN} (EN)	19,780	1,153	93,484
	DBP15K _{FR-EN} (FR)	19,661	903	105,998
	DBP15K _{FR-EN} (EN)	19,993	1,208	115,722
MMEL	MWE-EL	1,444	80	1,393
	MWD-EL	1,661	115	2,600

5 Experiments

5.1 Experimental Setup

As summarized in Table 1, we construct a comprehensive MMKG benchmark integrating three tasks: MMLP, MMEA and MMEL. For MMLP, we integrate widely-used datasets including YAGO15K, DB15K and FB15K-237 from MMKG[26], MKG-W and MKG-Y from AdaMF[53]. The MMEA task utilizes DBP15K[23, 36], which comprises three cross-lingual subsets—ZH-EN, JA-EN, and FR-EN—each containing approximately 15k aligned entity pairs and 40k triples. For MMEL, we construct two new small-scale datasets from Wikidata by extracting overlapping triplets with textual mentions: MWE-EL (constructed from the overlap of WikiMEL and MKG-W) and MWD-EL (constructed from WikiDiverse and MKG-W), which are derived from Wikidata[38] to ensure structural and multimodal consistency with aligned textual and visual mentions. Appendix A.1, A.2, A.3, and A.4 provide complete dataset statistical details, descriptions of baseline models, implementation configurations, and comprehensive evaluation protocols, respectively.

5.2 Multi-Task Inductive Inference Performance

As demonstrated in Table 2 and Table 3, MtaIMKG exhibits exceptional generalization capabilities under a challenging multi-task inductive learning paradigm. Our unified model is trained on two comprehensive dataset combinations: Combination A (YAGO15K

for MMLP, JA-EN for MMEA, MWD-EL for MMEL) and Combination B (DB15K for MMLP, FR-EN for MMEA, MWE-EL for MMEL). The evaluation strategy is designed to comprehensively assess the model’s capabilities. For MMEA and MMLP tasks, we conduct comparative analyses with SOTA baselines (Table 2), while using MMEL to demonstrate framework extensibility (Table 3).

On MMEA, MtaIMKG demonstrates impressive cross-lingual entity alignment capabilities. Particularly in the transfer from Combination A to FR-EN, our model achieves an MRR of 0.8449, significantly outperforming the best baseline MEAformer (0.1965) by a remarkable **329% relative improvement**. Even in more challenging cross-lingual scenarios such as from A to ZH-EN, the model maintains strong performance (0.6173 MRR), substantially exceeding MEAformer’s 0.2903. This validates that through multi-task training, our model learns language-agnostic entity representations, enabling robust inductive reasoning.

On MMLP, the results clearly reveal the limitations of conventional models. Even strong multimodal approaches like MoCi and MoMoK suffer severe performance degradation in inductive settings. For instance, in the A→DB15K transfer, MoCi’s MRR is only **31.1%** of our model’s performance, while MoMoK achieves merely **20.4%**. The fundamental reason lies in their dependence on static embeddings learned from training graphs, which fail to adapt to new graph distributions. In contrast, MtaIMKG’s dynamic task-aware embedding generation mechanism maintains strong performance when confronted with novel graph structures. Notably, while the structure-only inductive model ULTRA shows reasonable performance, our model still outperforms it by **71%** on the A→FB15K-237 task, highlighting the crucial importance of effectively integrating multimodal information for inductive reasoning.

Notably, this work presents the first integration of MMEL into a unified multi-task inductive reasoning framework. As shown in Table 3, MtaIMKG achieves strong performance in both transductive (MRR > 0.89) and inductive settings (MRR > 0.64). These results collectively demonstrate that our model’s task-aware routing network, dynamically integrates and routes multimodal representations to generate adaptive, task-conditioned embeddings for any input graph, effectively addressing feature, structural, and task heterogeneity in multimodal knowledge graphs. MtaIMKG is the first unified framework for multi-task inductive inference on MMKG. It marks a paradigm shift by seamlessly unifying MMLP, MMEA, and MMEL within a single model that achieves zero-shot generalization to unseen graphs, moving beyond task-specialized models.

5.3 Transductive Link Prediction Performance

To further validate the robustness and representational power of MtaIMKG, we evaluate our model on the conventional transductive link prediction task. As shown in Table 4, MtaIMKG achieves the best performance across all evaluation metrics on both MKG-W and MKG-Y datasets, outperforming all strong transductive-specialized MMLP baselines. On MKG-W, our model achieves an MRR of 0.3793, representing a **5.1% improvement** over the strongest baseline (MyGO, 0.3610). Notably, these superior results are achieved without any task-specific, graph-specific, or entity-specific embedding training. This demonstrates that our task-conditioned and graph-adaptive embeddings possess strong transferability, even under

Table 2: Multi-Task Inductive Performance of MtaIMKG vs. Baselines: Trained on Combination A (DB15K for MMLP, FR-EN for MMEA, MWE-EL for MMEL) and Combination B (YAGO15K for MMLP, JA-EN for MMEA, MWD-EL for MMEL).

Trained On A (YAGO15K, JA-EN, MWD-EL)							Trained On B (DB15K, FR-EN, MWE-EL)					
Task 1: MMEA												
Model	A to FR-EN			A to ZH-EN			B to JA-EN			B to ZH-EN		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
	Multimodal Entity Alignment Methods						Multimodal Entity Alignment Methods					
MCLEA _{COLING'22}	0.1115	0.0819	0.1623	0.2210	0.1727	0.3107	0.1536	0.1157	0.2209	0.4785	0.3859	0.6664
MEAformer _{ACMMM'23}	0.1965	0.1656	0.2473	0.2903	0.2501	0.3617	0.2446	0.2107	0.3043	0.3104	0.2725	0.3807
SimDiff _{KDD'24}	0.0520	0.0120	0.0830	0.0746	0.0407	0.1314	0.0501	0.0290	0.0837	0.0814	0.0473	0.1384
IBMEA _{ACMMM'24}	0.0394	0.0225	0.0645	0.1193	0.0799	0.1896	0.1086	0.0739	0.1673	0.1295	0.0887	0.2061
Ours	0.8449	0.8018	0.9204	0.6173	0.5564	0.7274	0.6874	0.6265	0.7948	0.7011	0.6475	0.7965
Task 2: MMLP												
Model	A to FB15K-237			A to DB15K			B to FB15K-237			B to YAGO15K		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
	Uni-modal Link Prediction Methods						Uni-modal Link Prediction Methods					
INGRAM _{ICML'23}	0.0130	0.0040	0.0260	0.0870	0.0430	0.1530	0.0980	0.0580	0.1750	0.0360	0.0190	0.0610
ULTRA _{ICLR'24}	0.1596	0.0916	0.2983	0.2048	0.1333	0.3493	0.2068	0.1252	0.3773	0.3582	0.2790	0.5087
HyRel _{ACMMM'24}	0.0610	0.0230	0.0930	0.1140	0.0830	0.1670	0.1240	0.0670	0.1810	0.0960	0.0410	0.1210
	Multimodal Link Prediction Methods						Multimodal Link Prediction Methods					
IMF _{WWW'23}	0.0210	0.0165	0.0297	0.0725	0.0630	0.0865	0.0187	0.0148	0.0259	0.0585	0.0482	0.0723
MoCi _{ACMMM'24}	0.0243	0.0179	0.0367	0.0796	0.0554	0.1363	0.0257	0.0187	0.0390	0.0919	0.0491	0.1804
MoMok _{ICLR'25}	0.0115	0.0082	0.0166	0.0523	0.0211	0.0847	0.0043	0.0027	0.0098	0.0123	0.0021	0.0167
Ours	0.2730	0.1847	0.4479	0.2560	0.1872	0.3872	0.2103	0.1319	0.3715	0.3632	0.2916	0.5003
Model	A to MKG-W			A to MKG-Y			B to MKG-W			B to MKG-Y		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
	Uni-modal Link Prediction Methods						Uni-modal Link Prediction Methods					
INGRAM _{ICML'23}	0.0060	0.0020	0.0130	0.0830	0.0510	0.1210	0.0860	0.0420	0.1060	0.0260	0.0120	0.0750
ULTRA _{ICLR'24}	0.2991	0.2319	0.4310	0.3513	0.3113	0.4299	0.3048	0.2421	0.4169	0.3272	0.2904	0.3919
HyRel _{ACMMM'24}	0.0230	0.0050	0.0410	0.1040	0.0630	0.1340	0.1040	0.0710	0.1360	0.0750	0.0310	0.0980
	Multimodal Link Prediction Methods						Multimodal Link Prediction Methods					
IMF _{WWW'23}	0.0959	0.0887	0.1035	0.0046	0.0021	0.0071	0.0840	0.0811	0.0867	0.0042	0.0017	0.0075
MoCi _{ACMMM'24}	0.1427	0.1023	0.2030	0.0206	0.0151	0.0263	0.1317	0.0991	0.1915	0.0181	0.0142	0.0250
MoMok _{ICLR'25}	0.0187	0.0094	0.0472	0.0087	0.0024	0.0145	0.0068	0.0029	0.0107	0.0104	0.0023	0.0195
Ours	0.3225	0.2599	0.4409	0.3549	0.3138	0.4327	0.3191	0.2546	0.4422	0.3363	0.2919	0.4185

Table 3: MMEL Performance of MtaIMKG.

Trained On A (YAGO15K, JA-EN, MWD-EL)				Trained On B (DB15K, FR-EN, MWE-EL)		
Task 3: MMEL						
Ours	MWD-EL			MWE-EL		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
(Transductive)	0.9232	0.8720	0.9840	0.8913	0.8622	0.9449
Ours	A to MWE-EL			B to MWD-EL		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
(Inductive)	0.6692	0.5760	0.8440	0.6436	0.5984	0.7362

evaluation protocols specifically designed for transductive models. The fact that a single model, trained under an inductive paradigm, can excel in both inductive and transductive settings underscores the generality and power of our approach. When combined with the multi-task inductive results, these findings not only validate the powerful representational capacity of MtaIMKG but also highlight its unique advantage as a unified framework that bridges the gap between inductive and transductive learning paradigms for MMKG.

5.4 Ablation Study

As shown in Table 5 (trained on A: YAGO15K, JA-EN, MWD-EL), MtaIMKG without introduced lightweight task heads still achieves notably competitive performance, demonstrating that the core modules alone effectively produce transferable, universal representations. Adding individual lightweight heads provides targeted gains on their respective tasks, while the full multi-head configuration produces the most balanced performance, confirming optimal synergy between shared representation learning and task specialization. Taken together, these results show that MtaIMKG balances task-specific refinement with cross-task knowledge sharing, providing a robust foundation for multi-task MMKG applications. Appendix B presents the modality ablation analysis results.

5.5 Efficiency Analysis

To assess the efficiency of our approach, we compare MtaIMKG against specialized single-task baselines in Table 6. Our unified model achieves remarkable parameter efficiency with only 10.4M

Table 4: Comprehensive Comparison of Transductive Link Prediction Performance on MKG-W and MKG-Y Datasets.

Model	MKG-W				MKG-Y			
	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10
<i>Uni-modal Link Prediction Methods</i>								
ComplEx (ICML'16)	0.2493	0.1909	0.2669	0.3673	0.2871	0.2226	0.3212	0.4093
RotatE (ICLR'16)	0.3367	0.2680	0.3668	0.4673	0.3495	0.2910	0.3835	0.4530
QuatE (NeurIPS'19)	0.3450	0.2894	0.3671	0.4664	0.3601	0.3053	0.3884	0.4368
DualE (AAAI'21)	0.3394	0.2755	0.3656	0.4609	0.3495	0.2977	0.3844	0.4312
<i>Multimodal Link Prediction Methods</i>								
VBKGC (KDD'22)	0.3061	0.2491	0.3301	0.4088	0.3704	0.3376	0.3875	0.4230
OTKGE (NeurIPS'22)	0.3436	0.2885	0.3625	0.4488	0.3551	0.3197	0.3718	0.4138
MoSE (EMNLP'22)	0.3334	0.2778	0.3394	0.4106	0.3628	0.3364	0.3747	0.4081
MMRNS (ACMMM'22)	0.3503	0.2859	0.3749	0.4747	0.3593	0.3053	0.3907	0.4547
QEB (ACMMM'23)	0.3238	0.2547	0.3506	0.4532	0.3437	0.2949	0.3695	0.4232
VISTA (EMNLP'23)	0.3291	0.2612	0.3538	0.4561	0.3045	0.2487	0.3239	0.4153
IMF (WWW'23)	0.3450	0.2877	0.3662	0.4544	0.3579	0.3295	0.3714	0.4063
AdaMF (COLING'24)	0.3427	0.2721	0.3786	0.4721	0.3806	0.3349	0.4044	0.4548
MoCi (ACMMM'24)	0.3581	0.3074	0.3801	0.4593	0.3887	0.3562	0.4041	0.4495
MyGO (AAAI'25)	0.3610	0.2978	0.3854	0.4775	0.3844	0.3501	0.3984	0.4419
K-ON (AAAI'25)	0.3583	0.2941	0.3732	0.4716	0.3583	0.3256	0.3734	0.4245
MoMok (ICLR'25)	0.3589	0.3038	0.3754	0.4613	0.3791	0.3509	0.3920	0.4320
Ours	0.3793	0.3171	0.4023	0.4959	0.3938	0.3574	0.4106	0.4592

Table 5: Impact of Task Heads on Multi-Task Inference.

LP	EA	EL	To FB15K-237		To FR-EN		To MWE-EL	
			MRR	Hit@1	MRR	Hit@1	MRR	Hit@1
✓	✓	✓	0.2517	0.1605	0.6134	0.5130	0.6242	0.5748
			0.2636	0.1759	0.6142	0.5126	<u>0.6451</u>	0.5984
✓	✓	✓	0.2772	0.1876	0.6291	0.5306	0.6325	<u>0.5827</u>
			0.2647	0.1768	0.8739	0.8393	0.6225	0.5742
✓	✓	✓	<u>0.2730</u>	<u>0.1847</u>	<u>0.8449</u>	<u>0.8018</u>	0.6692	0.5760

Table 6: Efficiency and Resource Consumption Comparison between the Proposed Model and Single-Task Baselines.

Model	GPU Memory	Parameters	Train Time	Inference Time	MMLP MRR	MMEA MRR	MMEL MRR
IMF	17GB	71.7M	7.1h	7.1h	0.0585	-	-
MoCi	29GB	39.3M	10.3h	10.3h	0.0919	-	-
SimDiff	28GB	0.4M	0.9h	0.9h	-	0.0814	-
IBMEA	34GB	2.37M	0.4h	0.4h	-	0.1295	-
Ours	37GB	10.4M	15.3h	0.1h	0.3632	0.7011	0.6692

parameters, **3.8× fewer** than individual specialized models like MoCi (39.3M). While multi-task training on datasets A (DB15K, FR-EN, MWE-EL) requires 15.3h, this represents a substantial reduction compared to sequential training of separate models. More importantly, MtaIMKG achieves superior inference efficiency at 0.1h, **71× faster** than MMLP baselines on YAGO15K, while also demonstrating strong zero-shot generalization for MMEA on ZH-EN and MMEL on MWD-EL. The 37GB memory footprint remains competitive while handling three tasks jointly, eliminating the need

for multiple specialized models. Crucially, MtaIMKG's amortized training enables zero-shot generalization across graphs through a single training session, whereas specialized models require retraining for each new scenario, making our approach ideal for dynamic environments with continuously emerging knowledge graphs. We provide a detailed analysis of key hyperparameters, including embedding size and adapter rank, in Appendix C.

6 Conclusion

In this study, we propose MtaIMKG, the first unified multi-task inductive reasoning framework for MMKG, which overcomes the limitations of traditional task-specialized paradigms. MtaIMKG demonstrates universal and transferable capabilities, effectively addressing inherent heterogeneity in multimodal knowledge graphs. Extensive experiments demonstrate that MtaIMKG achieves SOTA performance across multiple tasks while maintaining efficient inference. This work provides a scalable and generalizable solution for MMKG reasoning in dynamic environments, advancing robust and adaptable multimodal knowledge representation. Limitations and future directions are discussed in the Appendix D.

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A Experimental Setup Details

A.1 Dataset Details

This section provides comprehensive statistical summaries of the datasets used for each task in our study. Table 7 presents the five multimodal knowledge graph datasets employed for MMLP, detailing their scale in terms of entities, relations, and triplets, along with the standard data splits. Table 8 describes the three cross-lingual DBP15K subsets utilized for MMEA, showing the structural characteristics and alignment pairs for each language pair. Table 9 overviews the two Wikidata-derived datasets constructed for MMEL: MWE-EL (constructed from the overlap of WikiMEL and MKG-W) and MWD-EL (constructed from WikiDiverse and MKG-W), highlighting their composition and mention distributions across different splits. Complete numerical details for all datasets are available in the respective tables. These datasets preserve their original feature extraction methods and exhibit substantial heterogeneity across multiple dimensions. Structurally, they vary significantly in topological characteristics such as average degree, frequency of dense subgraphs, missing-edge ratio, and the prevalence of multi-hop dependencies. Domain-wise, they span diverse contexts including entertainment, sports, and academia, leading to vastly different relational semantics and entity distributions. This comprehensive heterogeneity in features, structures, and domains presents significant challenges for multi-task inductive reasoning, requiring models to overcome complex distribution shifts while maintaining robust performance across diverse scenarios.

A.2 Baselines

To evaluate our model’s performance in the Multi-Task Inductive Inference setting for MMKG, we select SOTA baselines from MMEA

and MMLP tasks. MMEL solely validates our model’s capability in handling this task. Under the inductive setting where models must handle unseen entities and relations without predefined entity/relation sets, we evaluate the following representative methods.

MMEA Methods. We include four advanced approaches: MCLEA [22] enhances cross-modal entity matching through contrastive learning; MEAformer[7] employs transformer networks to dynamically model multimodal interactions; SimDiff[20] utilizes diffusion-based augmentation for robust alignment in low-resource scenarios; and IBMEA[34] applies information bottleneck principles to reduce misleading multimodal information.

MMLP Methods. We comprehensively evaluate both recent multimodal SOTA methods and structurally-aware inductive models: IMF[21] emphasizes independent learning of modality-specific features through separate encoding pathways; MoCi[46] captures and integrates inter-entity modal semantics for enhanced representation learning; MoMoK[51] employs mixture-of-experts networks to disentangle modal representations for improved link prediction; along with INGRAM[14], HyRel[45], and Ultra[10], which leverage shared structural patterns of KGs to handle unknown entities and relations through structural inductive reasoning.

To further validate our model’s performance in conventional MMKG settings, we evaluate on the MMLP task using representative unimodal and multimodal models. Unimodal Models include ComplEx[37], RotatE[35], QuatE[48], and DualE[3]. Multimodal Models comprise VBKGC[54], OTKGE[4], MMRNS[44], QEB[40], MoSE[56], IMF[21], VISTA[18], AdaMF[53], MoCi[46], MyGO[52], K-ON[12], and MoMoK[51]. This selection provides comprehensive coverage of classic and SOTA approaches for thorough comparison.

A.3 Implementation Details

Experiments were conducted on NVIDIA RTX L40 GPUs with 48 GB of memory. The dimension of modality embeddings and the adapter rank size were set to 64 and 16, respectively, and the number of negative samples was set to 512 for link prediction. For parameter optimization, we adopted the Adam optimizer with its learning rate set to 5×10^{-4} . For baselines, we report both the originally published results and our re-implementations under the same protocol.

A.4 Evaluation Protocols

This section outlines the comprehensive evaluation framework adopted for assessing model performance across the three core tasks: MMLP, MMEA, and MMEL. Following established practices in knowledge graph completion [2], we employ two standard ranking-based metrics across all tasks:

- **Mean Reciprocal Rank (MRR):** Measures the average reciprocal rank of the correct entity across all test queries.
- **Hit@K:** Computes the proportion of test queries where the ground-truth entity appears in the top-K ranked candidates ($K = 1, 3, 10$).

MMLP Evaluation: We evaluate both head entity prediction ($(h, r, ?)$) and tail entity prediction ($(?, r, t)$). For each test triple, we generate candidate rankings by replacing the missing entity with all entities in the vocabulary and computing their scores.

MMEA Evaluation: For each entity in the source knowledge graph, we rank all entities in the target knowledge graph based

on similarity scores. Correctness is determined by whether the ground-truth aligned entity appears in the top-K candidates.

MMEL Evaluation: Given a mention, we rank all candidate entities from the knowledge graph. The evaluation assesses the model’s ability to correctly identify the referent entity among the ranked candidates.

Table 7: Overview of MMLP Datasets.

Datasets	#Ent	#Rel	#Triplets	Triplets Splits		
				Train	Valid	Test
YAGO15K [26]	15,283	32	122,886	86,020	12,289	24,577
DB15K [26]	14,777	279	99,028	69,319	9,903	19,806
FB15K-237 [26]	14,541	237	310,116	272,115	17,535	20,466
MKG-W [53]	15,000	169	42,746	34,196	4,276	4,274
MKG-Y [53]	15,000	28	26,638	21,310	2,665	2,663

Table 8: Overview of MMEA Datasets.

Dataset	KG	#Ent	#Rel	#Attr	#Triples	#EA Pairs	
						Train	Test
DBP15K _{ZH-EN}	ZH	19,388	1,701	8,111	70,414	4,500	10,500
	EN	19,572	1,323	7,173	95,142		
DBP15K _{JA-EN}	JA	19,814	1,299	5,882	77,214	4,500	10,500
	EN	19,780	1,153	6,066	93,484		
DBP15K _{FR-EN}	FR	19,661	903	4,547	105,998	4,500	10,500
	EN	19,993	1,208	6,422	115,722		

Table 9: Overview of MMEL Datasets.

Dataset	#Ent	#Rel	#Triples	#Mention	Mention Splits		
					Train	Valid	Test
MWE-EL	1,444	80	1,393	1,279	778	247	254
MWD-EL	1,661	115	2,600	1,244	746	248	250

B Modality Ablation Study

Within the multi-task inductive reasoning framework, we conducted modality ablation analysis using MMLP as an exemplar (Table 10), as MMLP’s structure-intensive nature better reveals the impact of modality fusion on fundamental reasoning capabilities. In contrast, MMEA and MMEL inherently rely on cross-modal semantic matching, where multimodal necessity has been well-established. Experimental results demonstrate that while structural information dominates MMLP reasoning with marginal gains from multimodal fusion, the consistent advantage of S+V+T configuration in cross-dataset inductive scenarios (A→MKG-W) indicates multimodal features effectively mitigate distribution shifts through visual and textual semantic consistency verification. This validates that despite varying modality sensitivity across individual tasks, unified multimodal integration remains a crucial design for ensuring model generalization across both tasks and knowledge graphs.

Table 10: Evaluation Results of Modality Combinations (Trained on A: YAGO15K, JA-EN, MWD-EL).

Model	Transductive			A to MKG-W		
	MRR	Hit@1	Hit@10	MRR	Hit@1	Hit@10
S	0.4488	0.3790	0.5783	0.3120	0.2543	0.4169
S+V	0.4494	0.3777	0.5776	0.3206	0.2578	0.4404
S+T	0.4500	0.3800	0.5797	0.3181	0.2546	0.4314
S+V+T	0.4539	0.3843	0.5801	0.3225	0.2599	0.4409

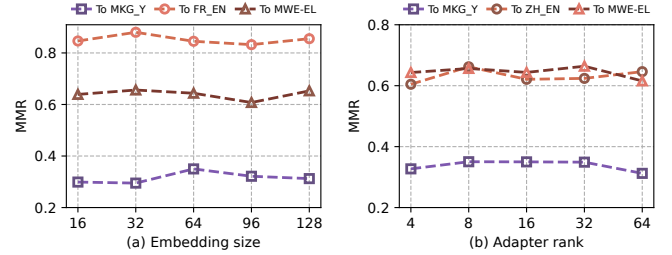


Figure 4: Impact of embedding size and Adapter rank size.

C Parameter Analysis

Fig. 4(a) illustrates the influence of embedding size on the performance of MtaIMKG. As shown, the embedding size plays an important role in model performance. It is worth noting, however, that larger embedding sizes do not always lead to better performance due to potential overfitting. Taking into account overall performance, efficiency, and the requirements of the inductive multi-task setting, we set the embedding size of MtaIMKG to 64. Additionally, Fig. 4(b) demonstrates the impact of the adapter rank, a numerical hyperparameter controlling the adapter complexity, on MtaIMKG’s performance. The results indicate that selecting an appropriate adapter rank is crucial for stable multi-task learning. After comprehensive evaluation across all tasks, we set the adapter rank to 16 as it achieves the stable overall multi-task performance, representing the optimal balance for MtaIMKG’s unified architecture.

D Limitations and Future Work

Due to the limitations of our current computational resources, our model has not yet been trained on multiple graphs. It remains unclear whether this approach could support models with enhanced generalization capabilities. In future work, we will explore how to scale the model’s parameter size and the number of training graphs to support models with stronger generalization abilities. Additionally, we plan to study the trade-offs between performance and efficiency in multi-graph training, particularly with respect to computational overhead and model effectiveness.